

USB-C 5V Power Board

Hardware Testing Procedure & Validation Record

KiCad 10 project: USB_C_5V_Power_Board

Purpose

This procedure validates the assembled board before it is used as a finished hardware project. Tests should be performed in the order listed. Record actual measured values in the result fields.

1. Visual Inspection

Inspect the PCB for damaged traces, solder bridges, loose components, incorrect orientations, damaged connectors, and contamination. Confirm USB-C connector, F1, D1, Q1, U1, LEDs, capacitors, resistors and output headers are correctly assembled.

Result: PASS / FAIL

2. Power-Off Short-Circuit Check

With power disconnected, use a multimeter in continuity/resistance mode. Check VBUS-to-GND, 5V-to-GND and 3.3V-to-GND. A brief changing reading may occur while capacitors charge; there must be no permanent short.

Result: PASS / FAIL

3. Controlled 5V Input Test

Use a known-good 5 V source. For first power-up, a current-limited bench supply is recommended, initially around 100–200 mA. Confirm the supply does not immediately enter current limit.

Result: VIN = ____ V; PASS / FAIL

4. 5V Output Test

Measure J2 between +5V and GND. Expected nominal output is approximately 5 V with only a small drop from the input under light load.

Result: V5 = ____ V; PASS / FAIL

5. 3.3V Output Test

Measure J3 between +3V3 and GND. Expected output is approximately 3.3 V; a practical acceptance target is about 3.3 V $\pm$ 0.1 V.

Result: V3V3 = ____ V; PASS / FAIL

6. LED Test

With the board powered, confirm D2 indicates the 5 V rail and D3 indicates the 3.3 V rail. If an LED is off, check orientation, resistor and voltage across the LED.

Result: D2 = ____; D3 = ____; PASS / FAIL

7. No-Load Current

With no external load connected to J2/J3, measure input current using a current-capable bench supply or suitable USB power meter.

Result: INOLOAD = ____ mA

8. 5V Load Test — 100 mA

Apply approximately 100 mA to the 5 V output. Equivalent resistor: $R = 5/0.1 = 50 \Omega$. Required load power is 0.5 W; use at least a 1 W resistor. Record output voltage and current.

Result: V5 = ____ V; I = ____ mA; PASS / FAIL

9. 3.3V Load Test — 100 mA

Apply approximately 100 mA to the 3.3 V output. Equivalent resistor: $R = 3.3/0.1 = 33 \Omega$. Load power is about 0.33 W; use at least a 1 W resistor.

Result: V3V3 = ____ V; I = ____ mA; PASS / FAIL

10. 3.3V Load Progression

After successful low-current testing, increase the intended 3.3 V load gradually: 100 mA → 200 mA → 300 mA. Record output voltage, input current and regulator temperature at each step.

Result: See load table below

11. AMS1117 Thermal Test

At the intended load, operate the board for approximately 10–15 minutes. Measure ambient and U1 temperature using an IR thermometer, thermocouple or thermal camera. Stop if abnormal heating or unstable output occurs.

Result: Tamb = ____ °C; TU1 = ____ °C; PASS / FAIL

12. Protection Verification

Visually and electrically verify the installed PTC, TVS and P-channel MOSFET. Do not intentionally short the powered input just to force the PTC to trip. Actual surge-clamping validation requires suitable test equipment.

Result: PASS / FAIL

13. Final ERC/DRC Confirmation

Confirm the design files remain clean after final changes: ERC = 0 violations and DRC = 0 violations. The existing intentional USB-C mounting-hole clearance exclusions may remain documented.

Result: ERC = ____; DRC = ____

Load Test Record

Load	Output voltage	Input current	U1 temperature	Status
100 mA	____ V	____ mA	____ °C	PASS / FAIL
200 mA	____ V	____ mA	____ °C	PASS / FAIL
300 mA	____ V	____ mA	____ °C	PASS / FAIL

Final Pass/Fail Summary

Visual inspection: ____ Short check: ____ 5V input: ____ 5V output: ____ 3.3V output: ____ LEDs: ____ Load tests: ____ Thermal test: ____ Protection: ____ ERC/DRC: ____

Safety and Test Notes

Use current-limited power for first power-up. Do not intentionally create a hard short on a powered board. Resistors used as loads must have adequate power rating and ventilation. The AMS1117 is the main thermal limitation of this design. The recommended documented continuous 3.3 V target is approximately 300 mA pending actual thermal validation.

Expected Design Targets

USB-C input: nominal 5 V. Protected 5 V output: approximately 5 V. Regulated output: approximately 3.3 V. USB-C operation is 5 V sink-only and does not implement USB Power Delivery.